

Z-Transform

$$x[n] \longleftrightarrow X(z)$$

$$\Rightarrow X(z) = \sum_{n=-\infty}^{\infty} x[n] z^{-n}$$

$$\Rightarrow z = re^{j\omega}$$

$\Rightarrow$ If $r=1$, then Z-transform $\equiv$ DTFT

Q $x[n] = a^n u[n]$. Compute Z-transform & its ROC.

$$X(z) = \sum_{n=-\infty}^{\infty} x[n] z^{-n}$$

$$= \sum_{n=-\infty}^{\infty} a^n u[n] z^{-n} = \sum_{n=0}^{\infty} a^n z^{-n}$$

$$= \sum_{n=0}^{\infty} (a z^{-1})^n = 1 + a z^{-1} + (a z^{-1})^2 + \dots$$

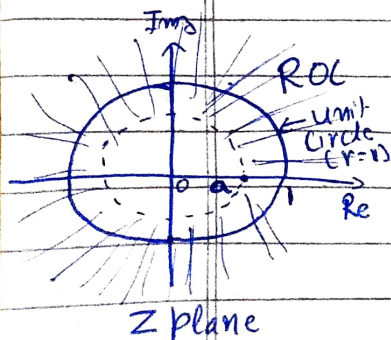
$$= \frac{1}{1 - a z^{-1}}$$

provided $|a z^{-1}| < 1$

$$\frac{|a|}{|z|} < 1$$

$$\rightarrow |a| < |z|$$

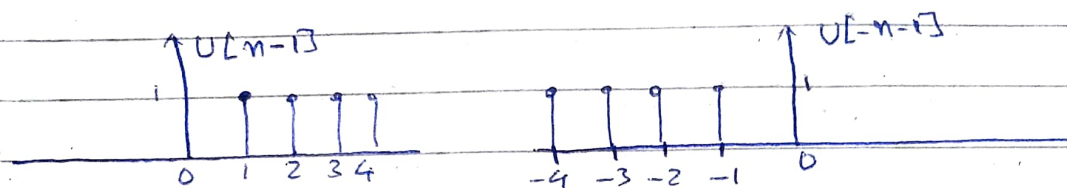
ROC
(Region of Convergence)



Q

$x[n] = -a^n u[-n-1]$ Find Z transform & its ROC

$u[n] \xrightarrow{\text{Delay}} u[n-1] \xrightarrow{\text{Reversal}} u[-n-1]$

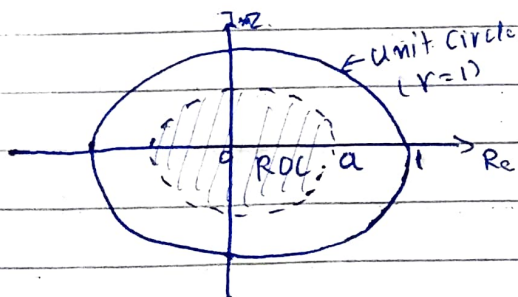


$$\begin{aligned}
 X(z) &= \sum_{n=-\infty}^{\infty} x[n] z^{-n} \\
 &= \sum_{n=-\infty}^{-1} -a^n z^{-n} = - \sum_{n=-\infty}^{-1} (az^{-1})^n \\
 &= - \left[\bar{a}^{-1}z + (\bar{a}^{-1}z)^2 + (\bar{a}^{-1}z)^3 + \dots \right] \\
 &= - \left[\frac{\bar{a}^{-1}z}{1 - \bar{a}^{-1}z} \right] = - \left[\frac{1}{az^{-1} - 1} \right] \\
 &= \frac{1}{1 - az^{-1}}
 \end{aligned}$$

ROC $\rightarrow |a^{-1}z| < 1$

$\frac{|z|}{|a|} < 1$

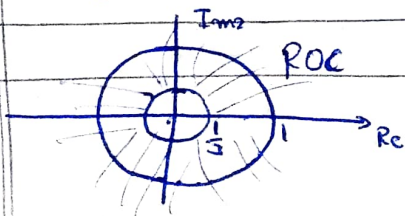
ROC $\rightarrow |z| < |a|$



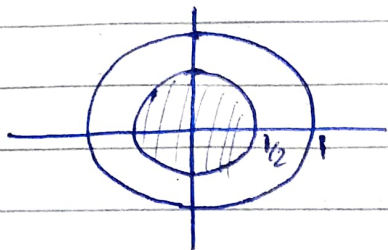
Q

$x[n] = \left(\frac{1}{3}\right)^n u[n] + \left(\frac{1}{2}\right)^n u[-n-1]$ Find Z transform & its ROC

$\left(\frac{1}{3}\right)^n u[n] \longleftrightarrow \frac{1}{1 - \left(\frac{1}{3}\right)z^{-1}} ; \text{ROC: } |z| > \frac{1}{3}$



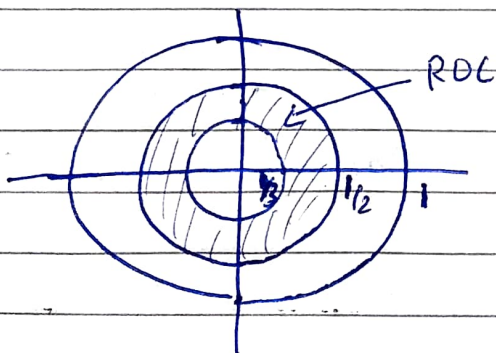
$$\left(\frac{1}{2}\right)^n U[-n-1] \longleftrightarrow -1 \frac{1}{1 - \frac{1}{2}z^{-1}} ; \text{ROC: } |z| < \frac{1}{2}$$



$$\left(\frac{1}{3}\right)^n U[n] + \left(\frac{1}{2}\right)^n U[-n-1] \longleftrightarrow \frac{1}{1 - \frac{1}{3}z^{-1}} - \frac{1}{1 - \frac{1}{2}z^{-1}}$$

ROC: Common Region of ROC of $\left(\frac{1}{3}\right)^n U[n]$ & $\left(\frac{1}{2}\right)^n U[-n-1]$

$$X(z) = \frac{z}{z - \frac{1}{3}} - \frac{z}{z - \frac{1}{2}} ; \text{ROC: } \frac{1}{3} < |z| < \frac{1}{2}$$



Q // $x[n] = a^{|n|}$; $a > 1$ Find z transform & its ROC

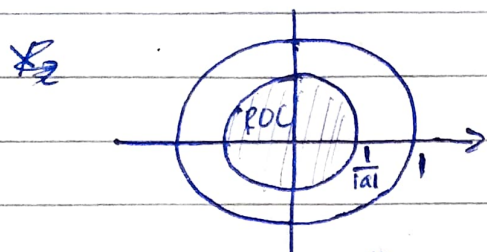
$$\begin{aligned} X(z) &= \sum_{n=-\infty}^{\infty} x[n] z^{-n} = \sum_{n=-\infty}^{-1} a^{-n} z^{-n} + \sum_{n=0}^{\infty} a^n z^{-n} \\ &= X_1(z) + X_2(z) \end{aligned}$$

$\uparrow$ $X_1(z)$ $\uparrow$ $X_2(z)$

$$X_1(z) = \frac{az}{1-az}$$

$$; \text{ROC: } |az| < 1$$

$$|z| < \frac{1}{|a|}$$



$$a > 1$$

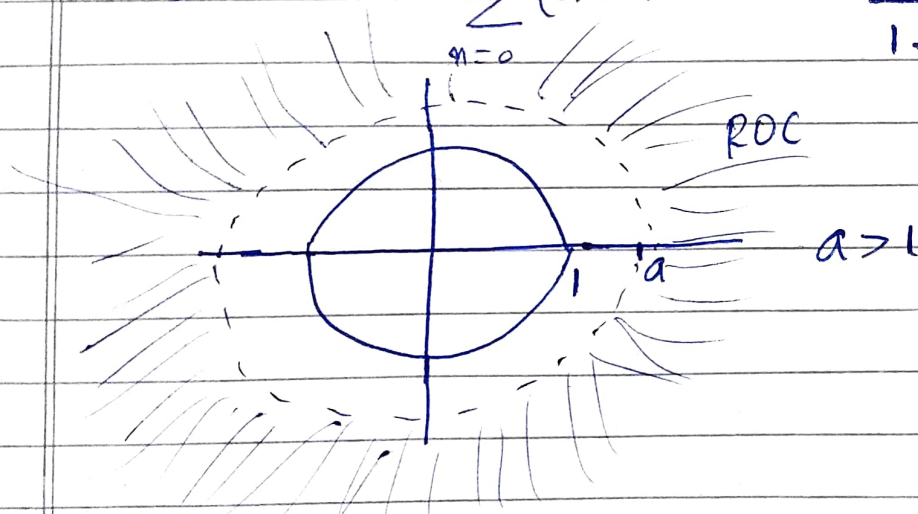
$$1 > \frac{1}{a}$$

$$X_2(z) = \sum_{n=0}^{\infty} (az^{-1})^n = \frac{1}{1-az^{-1}}$$

$$; \text{ROC: } |az^{-1}| < 1$$

$$\frac{|a|}{|z|} < 1$$

$$|z| > |a|$$



Since there is no common ROC (between ROC of X_1 & X_2), therefore Z transform does not exist for $a > 1$.

$X(z) \rightarrow$ Does not exist

⇒ Properties of Region of Convergence (ROC)

1. ROC consists of a ring in the z plane centered about origin
(Convergence is dependent on $r = |z|$ & not on w ($z = re^{jw}$))
$$\sum_{n=-\infty}^{\infty} |x(n)| r^{-n} < \infty$$

2. ROC does not contain any poles.

$$X(z) = \frac{N(z)}{D(z)}$$

Poles $\rightarrow$ Roots of $D(z)$

Zeros $\rightarrow$ Roots of $N(z)$

(@ Pole $X(z)$ is ∞ . So it cannot converge at Pole.)

3. If $x[n]$ is of finite duration, then ROC is the entire z plane except possibly $z=0$ &/or $z=\infty$.

Valid if $X(z)$ is Rational function of z

4. If $x[n]$ is a right sided sequence (i.e. $x[n]=0$ for $n < N_1 < \infty$), then ROC is of form: $|z| > r_{\max}$ or $\infty > |z| > r_{\max}$ where $r_{\max}$ equals the largest magnitude of any of the poles of $X(z)$.

5. If $x[n]$ is a left sided sequence (i.e. $x[n]=0$ for $n > N_2 > -\infty$), then ROC is of the form: $|z| < r_{\min}$ or $0 < |z| < r_{\min}$ where $r_{\min}$ is the smallest magnitude of any of the poles of $X(z)$.

6. If $x[n]$ is a two sided sequence (i.e. $x[n]$ is of infinite duration), then ROC is an annular ring such that $r_1 < |z| < r_2$, where r_1 & r_2 are the magnitude of two poles of $X(z)$.

Properties of Z transform

$$x_1[n] \longleftrightarrow X_1(z) \quad ; \text{ ROC: } R_1$$

$$x_2[n] \longleftrightarrow X_2(z) \quad ; \text{ ROC: } R_2$$

• Linearity property: $a x_1[n] + b x_2[n] \longleftrightarrow a X_1(z) + b X_2(z)$
 ROC: $R_1 \cap R_2$
 (Common region of R_1 & R_2).

• Time shifting: $x[n-n_0] \longleftrightarrow z^{-n_0} X(z)$
 ROC: ROC of $x[n]$ except for possible addition or deletion of the origin or infinity.

(Pole @ $z=0$ may be introduced or deleted due the term z^{-n_0}
 ROC cannot contain poles.)

• Scaling in Z Domain: $z_0^n x[n] \longleftrightarrow X\left(\frac{z}{z_0}\right) \quad ; \text{ ROC: } |z| \in R$

If $X(z)$ has pole at $z=a$, then $X\left(\frac{z}{z_0}\right)$ has pole at $z=z_0 a$
 $X(z) = \frac{z}{z-a}$, $X\left(\frac{z}{z_0}\right) = \frac{\frac{z}{z_0}}{\frac{z}{z_0}-a} = \frac{z}{z-z_0 a}$

$\therefore$ ROC is Scaled Version of R i.e. $|z| \in R$

- Time expansion: $x_{(10)}[n] = \begin{cases} x[n/10] & \text{if } n \text{ is multiple of } 10 \\ 0 & \text{"not a"} \end{cases}$
- $x_{(10)}[n] \longleftrightarrow X(z^{10})$; ROC: $R^{1/10}$

If $X(z)$ has pole at $z=a$, then $X(z^k)$ has pole at $z=a^{1/k}$

For $X(z) = \frac{z}{z-a}$; $\mathcal{Z}^{-1} X(z^k) = \frac{z^k}{z^k - a} \Rightarrow \text{Pole @ } z = a^{1/k}$

$$\therefore R_{OC} = R^{1/K}$$

- Conjugation: $x^*[n] \longleftrightarrow X^*(z^*)$; $\text{Roc} : R$

- Convolution Property: $x_1[n] * x_2[n] \longleftrightarrow X_1(z) X_2(z)$
ROC: $R_1 \cap R_2$

- Differentiation in Z Domain: $n x[n] \longleftrightarrow -z \frac{dx(z)}{dz}$
ROC: $\mathbb{R}$

- Initial value Theorem : $x[0] = \lim_{z \rightarrow \infty} X(z)$

Q If $x[n] \longleftrightarrow X(z)$ then find Z transform of $z_0^n x[n]$

$$\begin{aligned} Z[z_0^n x[n]] &= \sum_{n=-\infty}^{\infty} z_0^n x[n] z^{-n} = \sum_{n=-\infty}^{\infty} x[n] (z z_0^{-1})^{-n} \\ &= \sum_{n=-\infty}^{\infty} x[n] \left(\frac{z}{z_0}\right)^{-n} \\ &= X\left(\frac{z}{z_0}\right) \end{aligned}$$

Q If $x[n] \longleftrightarrow X(z)$, then find the z transform of $x[-n]$

$$Z[x[-n]] = \sum_{n=-\infty}^{\infty} x[-n] z^{-n}$$

$$\begin{aligned} Z[x[-n]] &= \sum_{n_1=-\infty}^{\infty} x[n_1] z^{n_1} \quad \text{Put } n_1 = -n \\ &= \sum_{n_1=-\infty}^{\infty} x[n_1] (z^{-1})^{-n_1} \\ &= X(z^{-1}) \\ &= X\left(\frac{1}{z}\right) \end{aligned} \quad \left[X(z) = \sum_{n=-\infty}^{\infty} x[n] z^{-n} \right]$$

Q Prove time expansion property of z transform

$$x_k[n] = \begin{cases} x[n/k] & ; n \rightarrow \text{multiple of } k \\ 0 & ; n \rightarrow \text{Not a multiple of } k \end{cases}$$

$$\begin{aligned} Z[x_k[n]] &= \sum_{n=-\infty}^{\infty} x_k[n] z^{-n} \\ &= \sum_{\substack{n=-\infty \\ \langle n = \text{multiple of } k \rangle}}^{\infty} x[n/k] z^{-n} \end{aligned}$$

$$\text{Let } \frac{n}{k} = n_1$$

$$\begin{aligned} &= \sum_{n_1=-\infty}^{\infty} x[n_1] z^{-(n_1 k)} \\ &= \sum_{n_1=-\infty}^{\infty} x[n_1] (z^k)^{-n_1} \\ &= X(z^k) \end{aligned}$$

Hence proved

Q If $x[n] \longleftrightarrow X(z)$ then find $Z[nx[n]]$

$$X(z) = \sum_{n=-\infty}^{\infty} x[n] z^{-n}$$

Diff. both sides w.r.t. z

$$\frac{dX(z)}{dz} = \sum_{n=-\infty}^{\infty} x[n] (-n) z^{-n-1}$$

$$= - \sum_{n=-\infty}^{\infty} n x[n] z^{-n-1}$$

$$= -z^{-1} \sum_{n=-\infty}^{\infty} n x[n] z^{-n}$$

$$-z \frac{dX(z)}{dz} = \sum_{n=-\infty}^{\infty} n x[n] z^{-n} = Z[nx[n]]$$

$$\therefore Z[nx[n]] = -z \frac{dX(z)}{dz}$$

Q Find Z transform of $\delta[n]$ & $u[n]$

$$Z[\delta[n]] = \sum_{n=-\infty}^{\infty} \delta[n] z^{-n}$$

$$= \delta[0] z^0$$

$$= 1$$

$$\therefore Z[\delta[n]] = 1$$

$$Z[u[n]] = \sum_{n=-\infty}^{\infty} u[n] z^{-n}$$

$$= \sum_{n=0}^{\infty} (1) z^{-n} = 1 + z^{-1} + z^{-2} + \dots$$

$$= \frac{1}{1 - z^{-1}}$$

$$; \text{ROC: } |z^{-1}| < 1$$

$$\frac{1}{|z|} < 1$$

$$|z| > 1$$

Q $X(z) = \frac{z}{z^2 - 3z + 1}$; ROC : $|z| < \frac{1}{2}$

Find $x[n]$.

Partial fraction expansion of $X(z)$

$$\frac{X(z)}{z} = \frac{1}{2z^2 - 3z + 1} = \frac{A}{2z(z-1)} = \frac{1}{(2z-1)(z-1)}$$

$$\frac{X(z)}{z} = \frac{A}{(2z-1)} + \frac{B}{(z-1)}$$

$$\frac{1}{2z^2 - 3z + 1} = \frac{A(z-1) + B(2z-1)}{(2z-1)(z-1)}$$

$$\frac{1}{2z^2 - 3z + 1} = \frac{(-A-B) + z(A+2B)}{(2z-1)(z-1)}$$

Comparing coefficient of constant term & z both sides

$$A + 2B = 0 \quad ; \quad -A - B = 1$$

$$A = -2B \quad ; \quad +2B - B = 1$$

$$A = -2 \quad ; \quad B = 1$$

$$\frac{X(z)}{z} = \frac{-2}{2z-1} + \frac{1}{z-1}$$

$$X(z) = -\frac{z}{z - \frac{1}{2}} + \frac{z}{z-1}$$

$$X(z) = -\frac{1}{1 - \frac{1}{2}z^{-1}} + \frac{1}{1 - z^{-1}}$$

Since ROC is $|z| < \frac{1}{2}$. Acc. to property of ROC, it is a left sided sequence.

$$-a^n u[-n-1] \longleftrightarrow \frac{1}{1 - az^{-1}}$$

$$\therefore x[n] = \left(\frac{1}{2}\right)^n u[-n-1] - (1)^n u[-n-1]$$

$$x[n] = \left[\left(\frac{1}{2}\right)^n - 1\right] u[-n-1]$$

Experiment :

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Q Using the method of long division method, find the inverse z transform of $X(z)$

$$X(z) = \frac{z}{(z-1)(z-2)} ; \text{ROC} : 1 < |z| < 2$$

Looking at ROC, $x[n]$ will be a two sided infinite sequence / signal.

First decompose into left & right sided sequence using partial fraction.

$$\frac{X(z)}{z} = \frac{1}{(z-1)(z-2)} = \frac{A}{(z-1)} + \frac{B}{(z-2)}$$

$$\frac{1}{(z-1)(z-2)} = \frac{A(z-2) + B(z-1)}{(z-1)(z-2)}$$

$$\frac{1}{(z-1)(z-2)} = \frac{(A+B)z + (-2A - B)}{(z-1)(z-2)}$$

Comparing coefficient both sides

$$A+B = 0 ; -2A - B = 1$$

$$A = -B ; 2B - B = 1$$

$$A = -1 ; B = 1$$

$$\frac{X(z)}{z} = \frac{-1}{(z-1)} + \frac{1}{(z-2)}$$

$$X(z) = \frac{-z}{z-1} + \frac{z}{z-2}$$

↑
 $X_1(z)$
Right Sided
Sequence
($1 < |z|$)

↑
 $X_2(z)$
Left Sided
Sequence
($|z| < 2$)

Long Division of $X_1(z)$

$$\begin{array}{r} -1 - z^{-1} - z^{-2} \dots \\ z-1 \overline{) \begin{array}{r} -z \\ -z+1 \\ \hline -1 \\ -1+z^{-1} \\ \hline -z^{-1} \\ -z^{-1}+z^{-2} \\ \hline -z^{-2} \\ \vdots \end{array}} \end{array}$$

$$X_1(z) = -1 - z^{-1} - z^{-2} \dots$$

$$x_1[n] = \{-1, -1, -1, \dots\}$$

$\uparrow$
 $x_1[0]$

Long Division of $X_2(z)$

$$\begin{array}{r} -\frac{1}{2}z - \frac{1}{4}z^2 - \frac{1}{8}z^3 \dots \\ -2+z \overline{) \begin{array}{r} z \\ z - \frac{1}{2}z^2 \\ \hline \frac{1}{2}z^2 \\ \frac{1}{2}z^2 - \frac{1}{4}z^3 \\ \hline \frac{1}{4}z^3 \\ \frac{1}{4}z^3 - \frac{1}{8}z^4 \\ \hline \frac{1}{8}z^4 \\ \vdots \end{array}} \end{array}$$

$$X_2(z) = -\frac{1}{2}z - \frac{1}{4}z^2 - \frac{1}{8}z^3 \dots$$

$$x_2[n] = \{\dots, -\frac{1}{8}, -\frac{1}{4}, -\frac{1}{2}\}$$

$\uparrow$
 $x_2[1]$

$$\therefore x[n] = x_1[n] + x_2[n]$$

$$x[n] = \{\dots, -\frac{1}{8}, -\frac{1}{4}, -\frac{1}{2}, -1, -1, -1, \dots\}$$

$\uparrow$
 $x[0]$

$\Rightarrow$ Taylor Series expansion of $\log(1+x)$

$$\log(1+x) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1} x^n}{n} ; |x| < 1$$

⊗ Find the inverse z transform of $X(z) = \log(1+az^{-1})$, $|z| > |a|$

$$\begin{aligned} X(z) &= \log(1+az^{-1}) \\ &= \sum_{n=1}^{\infty} \frac{(-1)^{n+1} (az^{-1})^n}{n} \end{aligned}$$

(Taylor Series expansion)

$$= \sum_{n=1}^{\infty} \frac{(-1)^{n+1} a^n}{n} z^{-n}$$

$$= \sum_{n=-\infty}^{\infty} \frac{(-1)^{n+1} a^n}{n} u[n-1] z^{-n}$$

$$\therefore x[n] = \frac{(-1)^{n+1} a^n}{n} u[n-1]$$

(By comparing the formula of z transform)

⇒ Analysis of LTI System using Z transform

$$x[n] \rightarrow \boxed{\text{LTI System}} \rightarrow y[n] = x[n] * h[n]$$

→ $h[n]$: Impulse response of the system

$$y[n] = x[n] * h[n]$$

Taking Z transform both sides

$$Y(z) = X(z) H(z)$$

Transfer function of system, $H(z) = \frac{Y(z)}{X(z)}$

⇒ Causal System : $\begin{cases} h[n] = 0 & \text{for } n < 0 \\ h[n] \neq 0 & \text{for } n \geq 0 \end{cases}$

∴ $h[n]$ is right sided

If $h[n] = \delta[n+1] \rightarrow$ Non Causal System

then $H(z) = z$ (∴ Causal System's $H(z)$ does not contain positive power of z)

$$H(z) = z = \frac{1}{\frac{1}{z}} \rightarrow \text{Non Causal}$$

Pole: $\frac{1}{z} = 0 \Rightarrow z = \frac{1}{0} = \infty$ (Pole at infinity)

∴ Causal system does not have pole at ∞ .

Hence ROC should include ∞ .

Condition for Causality:

1. ROC is exterior of circle outside the outermost pole.
2. With $H(z)$ expressed as ratio of polynomial in z , the order of numerator cannot be greater than the order of denominator.

⇒ Stability of LTI System ($h[n]$ is absolutely summable)

Condition for stable system: ROC of $H(z)$ includes the unit circle, $|z|=1$.

(In other words, a causal LTI system with rational $H(z)$ is stable if & only if all of the poles of $H(z)$ lie inside the unit circle)

Proof: BIBO Stability Criterion: Bounded input Bounded output
If input, $x[n]$ is bounded, then $|x[n]| < B$ for all n
(B is finite number)
↑ Bounded

Output of system, $y[n] = \sum_{k=-\infty}^{\infty} x[k] h[n-k]$

$$y[n] = \sum_{k=-\infty}^{\infty} x[k] h[n-k]$$

For bounded output, $|y[n]| < B$ for all n

$$|y[n]| = \sum_{k=-\infty}^{\infty} |h[k]| |x[n-k]|$$

If $x[n]$ is bounded then $x[n-k]$ is also bounded

For $|y[n]| < B$,

then $\sum_{k=-\infty}^{\infty} |h[k]|$ must be finite

$$\therefore \sum_{k=-\infty}^{\infty} |h[k]| < \infty$$

$$H(z) = \sum_{n=-\infty}^{\infty} h[n] z^{-n}$$

$$z = r e^{j\omega}$$

If $r=1$ then $z = e^{j\omega}$ ($|z|=|e^{j\omega}|=1$)

$$\begin{aligned} |H(z)| &= \sum_{n=-\infty}^{\infty} |h[n] e^{j\omega n}| = \sum_{n=-\infty}^{\infty} |h[n]| |e^{j\omega n}| \\ &= \sum_{n=-\infty}^{\infty} |h[n]| < \infty \end{aligned}$$

∴ For stable system, ROC of $H(z)$ must contain Unit Circle
($r=1$)
or $|z|=1$

Experiment :

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Q The output $y[n]$ of a discrete-time LTI system is found to be $2\left(\frac{1}{3}\right)^n u[n]$ when the input $x[n]$ is $u[n]$.

(a) Find the impulse response $h[n]$ of the system.

(b) Find the output $y[n]$ when the input $x[n]$ is $\left(\frac{1}{2}\right)^n u[n]$.

Sol. (a) $y[n] = 2\left(\frac{1}{3}\right)^n u[n]$; $x[n] = u[n]$

$$X(z) = \frac{1}{1-z^{-1}} ; Y(z) = \frac{2}{1-\frac{1}{3}z^{-1}}$$

$$H(z) = \frac{Y(z)}{X(z)} = \frac{2(1-z^{-1})}{\left(1-\frac{1}{3}z^{-1}\right)(1)} = \frac{2(z-1)}{\left(z-\frac{1}{3}\right)}$$

$$\frac{H(z)}{z} = \frac{2(z-1)}{z\left(z-\frac{1}{3}\right)} = \frac{A}{z} + \frac{B}{z-\frac{1}{3}}$$

$$\frac{2(z-1)}{z\left(z-\frac{1}{3}\right)} = \frac{A\left(z-\frac{1}{3}\right) + B(z)}{z\left(z-\frac{1}{3}\right)} = \frac{z(A+B) - \frac{1}{3}A}{z\left(z-\frac{1}{3}\right)}$$

Comparing coefficient both the sides

$$A+B=2 ; \frac{1}{3}A=2$$

$$A=6 ; B=-4$$

$$\frac{H(z)}{z} = \frac{6}{z} - \frac{4}{z-\frac{1}{3}}$$

$$H(z) = 6 - \frac{4z}{z-\frac{1}{3}}$$

Taking inverse Z transform

$$h[n] = 6\delta[n] - 4\left(\frac{1}{3}\right)^n u[n]$$

(Since $x[n]$ & $y[n]$ are right sided, so $h[n]$ will also be right sided)

$$(b) \quad Y(z) = H(z) X(z)$$

$$x[n] = \left(\frac{1}{2}\right)^n u[n] \longleftrightarrow X(z) = \frac{z}{z - \frac{1}{2}}$$

$$Y(z) = \left(6 - \frac{4z}{z - \frac{1}{3}}\right) \left(\frac{z}{z - \frac{1}{2}}\right)$$

$$Y(z) = \frac{(6z - 2 - 4z)z}{\left(z - \frac{1}{3}\right)\left(z - \frac{1}{2}\right)} = \frac{(2z - 2)z}{\left(z - \frac{1}{3}\right)\left(z - \frac{1}{2}\right)}$$

$$\frac{Y(z)}{z} = \frac{2z - 2}{\left(z - \frac{1}{3}\right)\left(z - \frac{1}{2}\right)} = \frac{A}{\left(z - \frac{1}{3}\right)} + \frac{B}{z - \frac{1}{2}}$$

$$\frac{2z - 2}{\left(z - \frac{1}{3}\right)\left(z - \frac{1}{2}\right)} = \frac{A\left(z - \frac{1}{2}\right) + B\left(z - \frac{1}{3}\right)}{\left(z - \frac{1}{3}\right)\left(z - \frac{1}{2}\right)} = \frac{(A+B)z + \left(-\frac{1}{2}A - \frac{1}{3}B\right)}{\left(z - \frac{1}{3}\right)\left(z - \frac{1}{2}\right)}$$

$$A + B = 2$$

$$; \quad -\frac{1}{2}A - \frac{1}{3}B = -2$$

$$A = 2 - B$$

$$; \quad 3A + 2B = 12$$

$$A = 8$$

$$3(2 - B) + 2B = 12$$

$$-B = 6$$

$$Y(z) = \frac{8z}{\left(z - \frac{1}{3}\right)} - \frac{6z}{\left(z - \frac{1}{2}\right)}$$

Taking inverse z transform

$$y[n] = 8\left(\frac{1}{3}\right)^n u[n] - 6\left(\frac{1}{2}\right)^n u[n]$$

Q A Causal discrete-time LTI system is described by

$$y[n] - \frac{3}{4}y[n-1] + \frac{1}{8}y[n-2] = x[n]$$

where $x[n]$ & $y[n]$ are the input & output of the system respectively.

(a) Determine the transfer function, $H(z)$

(b) Find the impulse response, $h[n]$ of the system

(c) Find the output of the system for step input, $u[n]$

$$y[n] \longleftrightarrow Y(z) \quad ; \quad x[n] \longleftrightarrow X(z)$$

$$y[n] - \frac{3}{4} y[n-1] + \frac{1}{8} y[n-2] = x[n]$$

Taking Z transform both sides

$$Y(z) - \frac{3}{4} z^{-1} Y(z) + \frac{1}{8} z^{-2} Y(z) = X(z)$$

$$Y(z) \left[1 - \frac{3}{4} z^{-1} + \frac{1}{8} z^{-2} \right] = X(z)$$

$$\frac{Y(z)}{X(z)} = \frac{1}{1 - \frac{3}{4} z^{-1} + \frac{1}{8} z^{-2}} = \frac{z^2}{z^2 - \frac{3}{4} z + \frac{1}{8}}$$

$$H(z) = \frac{z^2}{z^2 - \frac{3}{4} z + \frac{1}{8}} \quad \leftarrow \text{Transfer Function of System}$$

$$\frac{H(z)}{z} = \frac{z}{z^2 - \frac{3}{4} z + \frac{1}{8}} = \frac{8z}{8z^2 - 6z + 1} = \frac{8z}{(4z-1)(2z-1)}$$

$$\frac{H(z)}{z} = \frac{A}{4z-1} + \frac{B}{2z-1}$$

$$\frac{8z}{(4z-1)(2z-1)} = \frac{A(2z-1) + B(4z-1)}{(4z-1)(2z-1)} = \frac{(2A+4B)z + (-A-B)}{(4z-1)(2z-1)}$$

Comparing coefficients both the sides

$$2A + 4B = 8$$

$$-A - B = 0$$

$$2(-B) + 4B = 8$$

$$A = -B$$

$$B = 4$$

$$A = -4$$

$$H(z) = \frac{-4z}{4z-1} + \frac{4z}{2z-1} = \frac{-z}{z - \frac{1}{4}} + \frac{2z}{z - \frac{1}{2}}$$

Taking inverse Z transform

$$h[n] = -\left(\frac{1}{4}\right)^n u[n] + 2\left(\frac{1}{2}\right)^n u[n] \quad \leftarrow \text{Impulse response of System}$$

(Since System is Causal, so $h[n]$ is right sided sequence.)

$$(c) \quad x[n] = u[n]$$

$$X(z) = \frac{1}{1-z^{-1}} = \frac{z}{z-1}$$

$$Y(z) = H(z) X(z)$$

$$Y(z) = \left(\frac{z^2}{z^2 - \frac{3}{4}z + \frac{1}{8}} \right) \left(\frac{z}{z-1} \right)$$

$$\frac{Y(z)}{z} = \frac{8z^2}{(4z-1)(2z-1)(z-1)} = \frac{A}{4z-1} + \frac{B}{2z-1} + \frac{C}{z-1}$$

$$\frac{Y(z)}{z} = \frac{A(2z-1)(z-1) + B(4z-1)(z-1) + C(4z-1)(2z-1)}{(4z-1)(2z-1)(z-1)}$$

$$\frac{8z^2}{(4z-1)(2z-1)(z-1)} = \frac{z^2(2A+4B+8C) + z(-3A-5B-6C) + (A+B+C)}{(4z-1)(2z-1)(z-1)}$$

Comparing coefficient both sides

$2A + 4B + 8C = 8$	$-3A - 5B - 6C = 0$	$A + B + C = 0$
$A = 4 - 2B - 4C$	$-3(-4 + 2C) - 5(4 - 3C) - 6C = 0$	$4 - 2B - 4C + B + C = 0$
$A = 4 - 2(4 - 3C) - 4C$	$3C - 8 = 0$	$B = 4 - 3C$
$A = -4 + 2C$	$C = \frac{8}{3}$	$B = 4 - 3\left(\frac{8}{3}\right)$
$A = -4 + 2\left(\frac{8}{3}\right)$		$B = -4$
$A = \frac{4}{3}$		

$$Y(z) = \frac{\frac{4}{3}}{4z-1} - \frac{4z}{2z-1} + \frac{\frac{8}{3}}{z-1}$$

$$Y(z) = \frac{\frac{1}{3}}{z-\frac{1}{4}} - \frac{2z}{z-\frac{1}{2}} + \frac{\frac{8}{3}}{z-1}$$

Taking inverse Z transform

$$y[n] = \frac{1}{3} \left(\frac{1}{4}\right)^n u[n] - 2 \left(\frac{1}{2}\right)^n u[n] + \frac{8}{3} u[n]$$